Strand 2: Algebra

Sub-strand 1: Patterns and Relations

Lesson 1: Linear Relations, Gradient, and Equations of Straight Lines

1. Curriculum Standard and Overview

- **Strand**: 2 — Algebra
- **Sub-strand**: 1 — Patterns and Relations
- **Content Standard**: B9.2.1.1
- **Indicators**:
- **B9.2.1.1.1** — Construct tables of values for linear relations and graph them; find the equation of a line.
- **B9.2.1.1.2 / B9.2.1.1.3** — Use graphs of linear relations to find missing elements and solve equations.
- **Also links to** B9.1.4.1.4 (gradient as unit rate — reading graphs).
- **Mock-paper links**: P2 Q4(b) (gradient 7 through (–2, –3)); P1 Q20 (gradient –2 through (1,2) and (4,y)); P1 Q35–37 (steep / shallow / zero gradient on a walker's height graph).

2. Preamble — What Is a Relation?

A **relation** is a rule that turns an input x into an output y.

Example rule: `y = 2x + 1`. Input 3 → output `2(3) + 1 = 7`.

When the rule is a straight line, we call it a **linear relation**. Straight lines have two key properties:

1. **Gradient** (slope) — how steep the line is.
2. **Y-intercept** — where the line crosses the vertical axis.

3. Tables of Values

Pick x-values, apply the rule, record y.

Rule: `y = 2x - 3`

x	-2	-1	0	1	2	3
y = 2x-3	-7	-5	-3	-1	1	3

Plot the pairs `(x, y)` and join with a straight line.

4. Gradient

4.1 Definition

The **gradient** `m` measures rise over run:

`m = (y_2 - y_1)/(x_2 - x_1) = (change in y)/(change in x)`

Gradient	Line looks like		
`m > 0`	uphill left → right		
`m < 0`	downhill left → right		

`m = 0`	perfectly flat (horizontal)		
large $	m	$	very steep
small $	m	$	shallow

4.2 Reading a graph without a formula

On a distance–time or height–time graph:

- **Steepest positive** section = largest positive gradient (fastest climb).
- **Shallowest positive** section = smallest positive gradient (slow climb).
- **Zero gradient** = flat section (height not changing).
- **Negative** section = going down.

Mock P1 Q35–37: On a walker's height graph, you must compare sections A–H.

- Steepest positive → the section with the biggest height gain over the shortest horizontal run.
- Shallowest positive → small gain over a long run.
- Zero gradient → a flat horizontal piece.

(Always read the actual diagram in the paper; the letters A–H identify segments of the path.)

4.3 Example calculation

Line through (1, 2) and (4, –4). `m = (-4 - 2)/(4 - 1) = (-6)/(3) = -2`.

5. Equation of a Straight Line

5.1 Gradient–intercept form

`y = mx + c`

- `m` = gradient
- c = y-intercept (where `x = 0`)

5.2 Finding the equation when you know `m` and one point

Mock P2 Q4(b): gradient `m = 7`, passes through `P(-2, -3)`.

1. Start from `y = 7x + c`.
2. Substitute `x = -2`, `y = -3`:

`-3 = 7(-2) + c \Rightarrow -3 = -14 + c \Rightarrow c = 11`.

3. **Equation:** `y = 7x + 11`.

(ii) Point `R(k, 4)` lies on the line: `4 = 7k + 11 \Rightarrow 7k = -7 \Rightarrow k = -1`.

5.3 Two-point method

1. Find `m` from the two points.
2. Substitute one point into `y = mx + c` to find c.

6. Parallel and Perpendicular (preview)

- **Parallel** lines have **equal** gradients.
- **Perpendicular** lines have gradients `m_1 × m_2 = -1`.

7. Twenty Fully Worked Examples

Example 1 — Table of values

$y = 2x - 3$, $x = -2..3$ (see §3).

Example 2 — Table with negative coefficient

$y = -x + 5$:

x	0	1	2	3
y	5	4	3	2

Example 3 — Table with fraction

$y = (1)/(2)x - 3$ at $x = 0, 2, 4, 6$: $y = -3, -2, -1, 0$.

Example 4 — Gradient from two points

(1, 2) and (4, –4): $m = -2$.

Example 5 — Gradient from (2, 6) and (4, 12)

$m = (12-6)/(4-2) = 3$. *(Also constant of proportionality = 3.)*

Example 6 — Gradient from (1, 2) and (4, y) with $m = -2$ (Mock P1 Q20)

$m = (y - 2)/(4 - 1) = -2 \Rightarrow (y-2)/(3) = -2 \Rightarrow y - 2 = -6 \Rightarrow y = -4$. **Answer: D.**

Example 7 — Equation from gradient and point (Mock P2 Q4b)

$m = 7$ through $(–2, –3)$: $y = 7x + 11$.

Example 8 — Find k on the line (Mock P2 Q4b ii)

$R(k, 4)$ on $y = 7x + 11$: $k = -1$.

Example 9 — Equation from two points

(0, 1) and (2, 5): $m = 2$, $c = 1 \rightarrow y = 2x + 1$.

Example 10 — Identify gradient and intercept

$y = -3x + 4$: $m = -3$, $c = 4$.

Example 11 — Zero gradient

$y = 5$ is a horizontal line; $m = 0$.

Example 12 — Steepest of three lines

$y = 2x$, $y = 5x$, $y = -7x$: steepest positive is $y = 5x$; steepest overall (most steep incl. negative) is $y = -7x$.

Example 13 — Missing output

$y = 3x - 1$; find y when $x = 7$: $y = 20$.

Example 14 — Missing input

$y = 3x - 1$; find x when $y = 11$: $11 = 3x - 1 \Rightarrow x = 4$.

Example 15 — Graph reading (walker heights)

Section with the biggest rise over 1 unit of time = steepest positive gradient. *(In the mock, compare sections A–H visually.)*

Example 16 — Flat section

If height stays constant over an interval, gradient = 0 (Mock P1 Q37 style).

Example 17 — Real-life rate as gradient

Cost = $5 \times \text{items}$. Gradient = GH■ 5 per item (unit rate).

Example 18 — Distance–time gradient

Distance–time graph gradient = **speed**. Steeper = faster.

Example 19 — Perpendicular preview

$m_1 = 2$; perpendicular $m_2 = -(1)/(2)$.

Example 20 — Parallel lines

Line $y = 4x + 1$; parallel through (0, 9): $y = 4x + 9$.

8. Comprehensive Exercise Sets (40 Questions)

Authentic mock items included in full. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Tables of values

1. Complete the table for $y = 2x - 3$, $x = -2..3$.
2. Complete the table for $y = -x + 5$, $x = 0..3$.
3. Complete the table for $y = (1)/(2)x - 3$, $x = 0, 2, 4, 6$.
4. For $y = 3x - 1$, find y when $x = 7$.
5. **Spiral review:** Share GH■ 720 in the ratio 2 : 3.

Exercise 2 — Gradient from two points (includes mock)

1. Find the gradient of the line through (1, 2) and (4, –4).
2. **[Mock 2026 P1 Q20]** A line passes through (1, 2) and (4, y). What is y if the gradient is –2?

A. 4 B. –2 C. 0 D. –4

3. Find the gradient through (2, 6) and (4, 12).
4. Find the gradient through (0, 1) and (2, 5).
5. **Spiral review:** 9 men finish work in 51 days. Days for 17 men?

Exercise 3 — Equation of a line (includes mock)

1. **[Mock 2026 P2 Q4(b)]** A straight line has gradient 7 and passes through P(–2, –3). Find (i) the equation of the line; (ii) the value of k if R(k, 4) lies on the line.
2. Find the equation of the line through (0, 1) and (2, 5).

3. Write down the gradient and y-intercept of $y = -3x + 4$.
4. Find the equation of the line through (0, 9) parallel to $y = 4x + 1$.
5. **Spiral review:** Evaluate $(2)/(3) \times (1)/(4) + (2)/(5) \div (14)/(15)$.

Exercise 4 — Reading graphs (includes mock)

1. **[Mock 2026 P1 Q35]** On the walker's height graph, which section shows the **steepest positive** gradient?

A. Section A B. Section B C. Section H D. Section F

2. **[Mock 2026 P1 Q36]** Which section shows the **shallowest positive** gradient?

A. Section A B. Section B C. Section F D. Section G

3. **[Mock 2026 P1 Q37]** In which section is the gradient **zero**?

A. Section A B. Section B C. Section D D. Section E

4. On a distance–time graph, what does the gradient represent?
5. **Spiral review:** A van travels 154 km in $1(3)/(4)$ hours. Find its speed.

Exercise 5 — Missing elements

1. For $y = 3x - 1$, find x when $y = 11$.
2. The line $y = 2x + c$ passes through (3, 10). Find c.
3. A line has gradient 5 and y-intercept –2. Write its equation.
4. **Spiral review:** $124.3 + 0.275 + 74.06$ to 1 d.p.
5. **Spiral review:** Classify 0 under ■, ■, ■, ■.

Exercise 6 — Slope as unit rate

1. Cost = $5 \times \text{items}$. What is the unit rate?
2. On a distance–time graph, a car covers 80 km in 2 hours. What is the gradient?
3. Which is faster: a gradient of 40 (km/h) or 25 (km/h)?
4. **Spiral review:** $(0.07 \times 0.02) \div 14$.
5. **Spiral review:** Find n if $(n)/(100) = 10.5$.

Exercise 7 — Parallel and perpendicular

1. What is the gradient of a line parallel to $y = 4x + 1$?
2. What is the gradient of a line perpendicular to $y = 2x + 3$?
3. Find the equation of the line through (0, 9) parallel to $y = 4x + 1$.
4. **Spiral review:** $(-3)^{(x-2)} = -243$. Find x.
5. **Spiral review:** Write 31 days in minutes, in standard form.

Exercise 8 — Mixed linear challenge

1. A line has equation $y = 7x + 11$. Find y when $x = -1$.
2. Does the point (2, 25) lie on $y = 7x + 11$? Justify.
3. The points (1, 3) and (3, 7) lie on a line. Find its equation.
4. **Spiral review:** 28 sell mackerel, 18 tilapia, 13 both, 36 total. How many sell neither?
5. **Spiral review:** Order $(3)/(8), (1)/(2), (5)/(16)$.

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. $y = -7, -5, -3, -1, 1, 3$.
2. $y = 5, 4, 3, 2$.

3. $y = -3, -2, -1, 0$.
4. $y = 20$.
5. GH■ 288 and GH■ 432.

Solutions to Exercise 2

1. $m = (-4-2)/(4-1) = -2$.
2. $(y-2)/(3) = -2 \Rightarrow y = -4 \rightarrow$ **D**.
3. $m = 3$.
4. $m = (5-1)/(2-0) = 2$.
5. 27 days.

Solutions to Exercise 3

1. (i) $c = 11 \rightarrow y = 7x + 11$. (ii) $4 = 7k + 11 \Rightarrow k = -1$.
2. $m = 2$, $c = 1 \rightarrow y = 2x + 1$.
3. $m = -3$, $c = 4$.
4. $y = 4x + 9$.
5. $(25)/(42)$.

Solutions to Exercise 4

1. Compare rise/run of each section in the mock diagram → **B** (Section B in the mock key).
2. Smallest positive rise/run → **C** (Section F).
3. Flat horizontal piece → **C** (Section D).
4. **Speed** (km/h).
5. $154 \div 1.75 = 88$ km/h.

Solutions to Exercise 5

1. $11 = 3x - 1 \Rightarrow x = 4$.
2. $10 = 6 + c \Rightarrow c = 4$.
3. $y = 5x - 2$.
4. 198.6.
5. ■, ■, ■.

Solutions to Exercise 6

1. **GH■ 5 per item**.
2. $80 \div 2 = 40$ km/h.
3. **40 km/h is faster**.
4. 0.0001.
5. $n = 1050$.

Solutions to Exercise 7

1. 4.
2. $m_1 m_2 = -1 \Rightarrow m_2 = -(1)/(2)$.
3. $y = 4x + 9$.
4. $x = 7$.
5. 4.464×10^4 minutes.

Solutions to Exercise 8

1. $y = 7(-1) + 11 = 4$.
2. $7(2) + 11 = 25 \rightarrow$ **yes, it lies on the line**.

3. $m = 2, c = 1 \rightarrow y = 2x + 1$.
4. 3.
5. $(5)/(16) < (3)/(8) < (1)/(2)$.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P2 Q4(b)	$m=7$ through $(-2,-3)$; $R(k,4)$	$y = 7x + 11$; $k = -1$
P1 Q20	$(1,2)$ and $(4,y)$, $m = -2$	D — $y = -4$
P1 Q35	Steepest positive gradient section	B (Section B in key)
P1 Q36	Shallowest positive gradient section	C (Section F)
P1 Q37	Zero gradient section	C (Section D)

11. Lesson Summary

Skill	Rule
Gradient	$m = (y_2 - y_1)/(x_2 - x_1)$
Equation	$y = mx + c$
Flat line	$m = 0$
Uphill / downhill	$m > 0$ / $m < 0$
Parallel	equal gradients
Perpendicular	$m_1 m_2 = -1$

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*